



The order of the product of two elements

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Abstract We prove that in a finite group G , $|ab| = |a||b|$ if and only if $ab = ba$ for all elements a, b of G whose orders are factor of the m -th power of different primes, where m is a given integer.

Keywords Finite group · Nilpotent group · Elements of co-prime order · Product

AMS Subject Classifications: 20B05 · 20D15

All groups in this paper are finite. For an element a of a group G , $|a|$ and a^G mean the order of a and the set of all conjugate elements of a in G respectively. Let p be a prime number, then $O_p(G)$ means the largest normal p -subgroup of G . The other notations and terminology are standard, as in [7, 9].

Let G be a finite group. Let $\Upsilon_\infty(G)$, $\Upsilon_k(G)$ and $D_k(G)$ denote the nilpotent residual, the k -th term of the lower central series and the k -th term of the lower Fitting series of G , respectively. $\Upsilon_k(G)$ and $D_k(G)$ are defined by $D_0(G) = \Upsilon_1(G) = G$ and $D_{i+1}(G) = \Upsilon_\infty(D_{i-1})$ and $\Upsilon_{i+1}(G) = [\Upsilon_i(G), G]$ for $i = 0, 1, 2, \dots$. The coprime commutator δ_k^* has been introduced in [4] by Shumyatsky. Every element of G is a δ_0^* -commutator, and the element g is a δ_k^* -commutator for $k > 0$ if there exist δ_{k-1}^* -commutators a and b with $(|a|, |b|) = 1$ such that $g = [a, b]$. It was shown in [4] that for every $k \geq 0$ the subgroup generated by δ_k^* -commutator is precisely $D_k(G)$.

The order of the product of two elements of G of co-prime orders is closely related to the structure of G . The special case which is the product of two elements of G of co-prime orders k and m has order km has attracted some attention recently. The following sufficient condition for nilpotency of a group G was discovered by Benjamin Baumslag and James Wiegold in [1].

Theorem 0.1 *Let G be a finite group in which $|ab| = |a||b|$ whenever the elements a, b have coprime orders. Then G is nilpotent.*

Later, R. Bastos and P. Shumyatsky [2], Victor S. Monakhov [5], R. Bastos, C. Monetta and P. Shumyatsky [3], A. F. Andrade and A. C. Dantas in [6] proved that G' , $\Upsilon_\infty(G)$ and $D_k(G)$ are nilpotent under some conditions. In this paper, we establish the relation between the commutativity and the order of the product of two elements of G of co-prime prime power orders. The main results are Theorem 0.1 and 0.3.

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Theorem 0.2 *Let G be a finite group. Then G is nilpotent if and only if $|ab| = |a||b|$ when a, b are the elements of co-prime prime power orders.*

Proof Since nilpotent groups are the direct product of its distinct Sylow subgroups, necessity is clear. Hence we just need to prove sufficiency. Assume that G is not a nilpotent group. Then there is a non-nilpotent subgroup H of minimal order in G such that every subgroup of H is nilpotent. By [7, Kapitel III, 5.2 Satz], $H = PQ$, $P \trianglelefteq H$, $\exp(P)$ is a prime or 4 and Q is cyclic, where P and Q are different Sylow subgroups of H . Thus there are elements $x \in P$ and $y \in Q$ such that $|xy| \neq |x||y|$, a contradiction. Hence G is nilpotent. $\square$

Let $\pi(G)$ denote the set of all prime divisors of $|G|$, $\Omega_{p^m}(G) = \{x \in G \mid x^{p^m} = 1\}$ for $p \in \pi(G)$ and $m \geq 2$, $\Omega_p(G) = \{x \in G \mid x^p = 1\}$ if $p > 2$ and $\Omega_2(G) = \Omega_{2^2}(G)$ if $p = 2$.

Theorem 0.3 *Let G be a finite group and m a given positive integer. Then the following statements are equivalent.*

- ($\mathcal{H}$) *For any different primes p and q , if $a \in \Omega_{p^m}(G)$ and $b \in \Omega_{q^m}(G)$, then $|ab| = |a||b|$.*
- ($\mathcal{A}$) *For any different primes p and q , if $a \in \Omega_{p^m}(G)$ and $b \in \Omega_{q^m}(G)$, then $ab = ba$.*

Proof If G satisfies the condition ($\mathcal{A}$), it is clear that G satisfies the condition ($\mathcal{H}$). Hence we need only to prove that ($\mathcal{H}$) implies ($\mathcal{A}$).

Let G be a counterexample of minimal order. Then there are two elements $a \in \Omega_{p^m}(G)$ and $b \in \Omega_{q^m}(G)$ such that $G = \langle a, b \rangle$ and $|ab| = |a||b|$ but $ab \neq ba$.

Assume that $\langle a^G \rangle < G$ or $\langle b^G \rangle < G$. Without loss of generality, let $\langle a^G \rangle < G$. Then $G = \langle a^G \rangle \langle b \rangle$. Assume that there is a prime $r \in \pi(\langle a^G \rangle)$ with $(p, r) = 1$. Let $R \in \text{Syl}_r(\langle a^G \rangle)$. Then $xa^g = a^g x$ for any $x \in \Omega_{p^m}(R)$ and $g \in G$ and so $\Omega_{p^m}(R) \subseteq Z(\langle a^G \rangle)$. By [8, Theorem 1], $\langle a^G \rangle$ is r -nilpotent. Let T be a normal r -complement in $\langle a^G \rangle$. Then $T \trianglelefteq G$ and $a \in T$. Thus $T \langle b \rangle < G$, contradiction. Hence $\langle a^G \rangle$ is a p -group and so $\langle a^G \rangle = P \in \text{Syl}_p(G)$. We have $G = P \langle b \rangle$. Since G is not nilpotent, there is a non-nilpotent subgroup H_0 of minimal order in G such that every proper subgroup of H_0 is nilpotent. It is evident that pq divides $|H_0|$. Assume that Q_0 is a Sylow q -subgroup of H_0 . Then, by Sylow's Theorem, there is an element $x \in G$ such that $Q_0^x \leq \langle b \rangle$. Let $H = H_0^x$, then we have $H = P_0 \langle b^k \rangle$ for some integer k , where P_0 is a subgroup of P . By [7, Kapitel III, 5.2 Satz], $P_0 = \Omega_p(P_0)$. If $H \neq G$, then by the choice of G , $xb^k = b^k x$ for any $x \in P_0$, a contradiction. Hence $G = H$, that is, $P_0 = P$ and $k = 1$. It is clear that $[a, b] \in P$ and $[a, b] = p$ or 4 . Since $[a, b]b^{-1} = a^{-1}b^{-1}abb^{-1} = b^{-a}$, we have $[a, b]b^{-1} = q \neq pq$, a contradiction. This contradiction shows $\langle a^G \rangle = \langle b^G \rangle = G$.

Since $\langle a \rangle G' = \langle a^G \rangle G' = \langle b^G \rangle G' = \langle b \rangle G'$ and $(|a|, |b|) = 1$, we have $G' = G$.

Assume that $x \in G$ and $|x| = 2$. For any $g \in G$, let $y = x^g$ and $H = \langle x, y \rangle$. Then $H = \langle x, xy \rangle$, H is a dihedral group and

$$(xy)^x = yx^{-1} = y^{-1}x^{-1} = (xy)^{-1},$$

hence $\langle xy \rangle \trianglelefteq H$, $H = \langle xy \rangle \cdot \langle x \rangle$. If $|x|$ is not a power of 2, then there is a positive integer k such that $|(xy)^k|$ is an odd prime. Thus by the condition ($\mathcal{H}$) we have $|x(xy)^k| = |x||x(y)^k|$. On the other hand, since

$$(x(xy)^k)^2 = x(xy)^k x(xy)^k = x^{-1}(xy)^k x(xy)^k = (xy)^{-k}(xy)^k = 1,$$

we have $|x(xy)^k| = 2$, a contradiction. Hence H is a 2-group. By the Baer-Suzuki Theorem (see [9, 39.6, p.204]), $x \in O_2(G)$. Without loss of generality, we may assume that $|a|$ is a power of an odd prime p . Since $G' = G$, we have $O_2(G) \langle a^g \rangle < G$ for any $g \in G$, thus $xa^g = a^g x$ by the choice of G . Since $\langle a^G \rangle = G$, we have $x \in Z(G)$.

Assume that $x \in G$ and $|x| = 4$. For any $g \in G$, let $y = x^g$ and $H = \langle x, y \rangle = \langle x, xy \rangle$. Since $|y^2| = 2$, we have $y^2 \in Z(G)$, $y^2 = (x^g)^2 = (x^2)^g = x^2$, and

$$xy = x \cdot x^4 \cdot y = x \cdot x^2 \cdot y^2 \cdot y = x^3 y^3 = x^{-1} y^{-1} = (yx)^{-1}.$$

Hence

$$(xy)^x = yx = (xy)^{-1}, (xy)^y = y^{-1}xy^2 = yx = (xy)^{-1},$$

thus $\langle xy \rangle \trianglelefteq H$, $H = \langle xy \rangle \cdot \langle x \rangle$. If $|x|$ is not a power of 2, then there is a positive integer k such that $|(xy)^k|$ is an odd prime. Thus by the condition ($\mathcal{H}$) we have $|x(xy)^k| = |x||x(y)^k|$. On the other hand, we have



$$(x(xy)^k)^2 = x(xy)^k x(xy)^k = x^2 x^{-1} (xy)^k x(xy)^k = x^2 (xy)^{-k} (xy)^k = x^2,$$

that is, $x(xy)^k$ is an element of order 4, a contradiction. Hence H is a 2-group. By the Baer-Suzuki Theorem (see [9, 39.6, p.204]), $x \in O_2(G)$. Similar to the previous proof, we have $x \in Z(G)$.

Summing the previous proof, we have proved that every element of order 2 or 4 is contained in $Z(G)$. By [8, Theorem 1], G is 2-nilpotent and so G is solvable by the Feit-Thompson Theorem (see [9, p.259] if necessary), contrary to $G' = G$. $\square$

Remark 0.4 It is clear that Theorem 0.2 is corollary of Theorem 0.3.

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